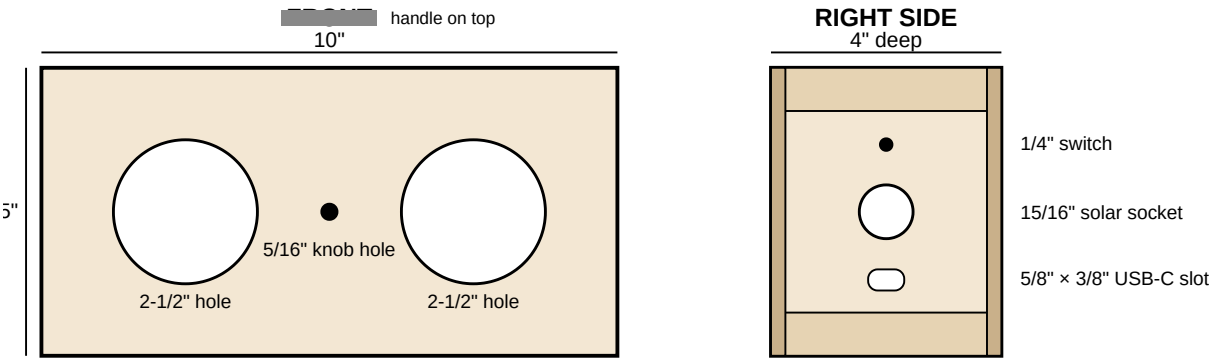
SURVIVAL RADIO · WOODEN BOX

Simple box plans. One 1×4 board, a scrap of 1/4" plywood, and an afternoon. Outside size 10" wide × 5" tall × 4" deep.



Front and back are 1/4" plywood. Top, bottom and sides are 1×4 pine (actually 3/4" × 3-1/2").

Sides sit between the top and bottom. The front is glued and nailed on; the back is screwed on so you can open it.

Full-size drilling templates for the front and right side are on pages 3 and 4.

Cut list

Part	Material	Size	Qty
Top and bottom	1×4 pine	10" long	2
Sides	1×4 pine	3-1/2" long	2
Front and back	1/4" plywood	10" × 5"	2

Buy one 3 ft 1×4 and a 12" × 12" (or larger) piece of 1/4" plywood.

Hardware

Wood glue	8 × 1-1/4" #6 wood screws (frame)
1" brad nails, or more glue and clamps (front)	4 × 5/8" #4 wood screws (back)
A drawer pull with screws, or 2 ft of 1/4" rope (handle)	Sandpaper (120 and 220); optional oil or spar urethane

Drill bits

2-1/2" hole saw · 15/16" spade or Forstner bit · 5/16" · 3/8" · 1/4" · 7/64" pilot bit · a small chisel or file for the USB-C slot.

Build it

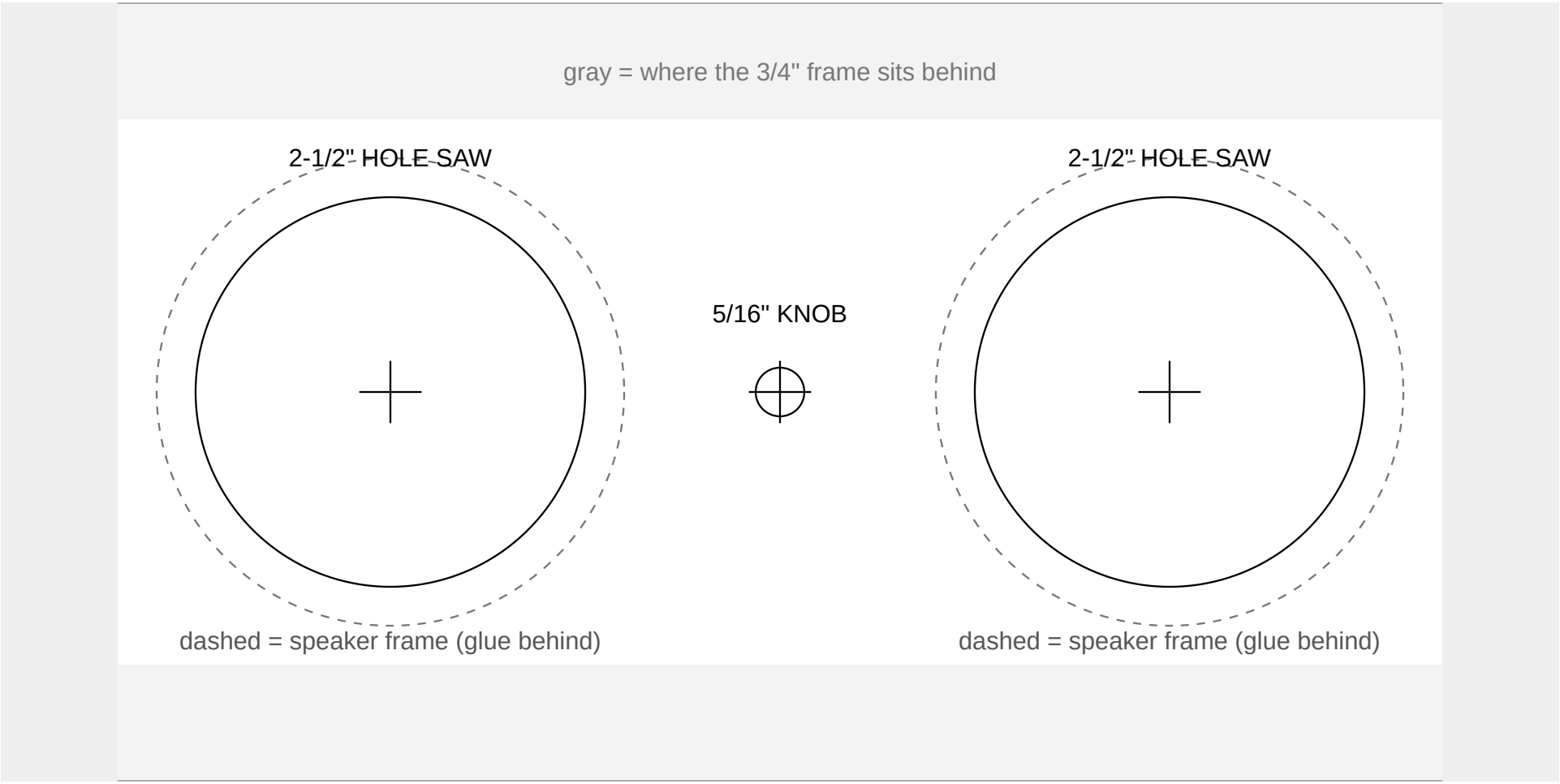
1. Cut the pieces from the cut list. Sand the cut ends smooth.
2. Tape the page 3 template to the front plywood. Drill the two 2-1/2" speaker holes and the 5/16" knob hole. Drill from the good face onto a scrap board to stop splintering.
3. Tape the page 4 template to the outside of the right side piece. Drill the 15/16" solar socket hole and the 1/4" switch hole. For the USB-C slot, drill two 3/8" holes side by side and clean out between them with a chisel or file.

4. Build the frame: put the sides between the top and bottom, glue, pre-drill with the 7/64" bit, and screw through the top and bottom into the sides (2 screws per joint). Check it's square.
5. Glue and nail the front onto the frame, with the speaker holes centered. Let the glue dry.
6. Lay the back panel on, pre-drill 3/8" in from each corner, and screw it on with the four #4 screws. That's your door for the electronics.
7. Drill a few 1/4" holes in the back panel for air, and attach the handle on top (drawer pull, or two 3/8" holes and a rope knotted inside).
8. Sand, then finish with oil, or spar urethane if it will live outside. Keep finish off the speaker cones.
9. Mount the parts (see the guide on edensgate.art): speakers glued behind their holes, the knob through its hole held by the encoder's nut, the solar socket and switch in the side, the charger board lined up with the USB-C slot, and the battery on the floor held with a velcro strap. Boards go on the inside of the back panel on M2.5 standoffs.

Survival Radio · free plans from Eden's Gate (edensgate.art/tools/survival-radio). Free to use, change and share.

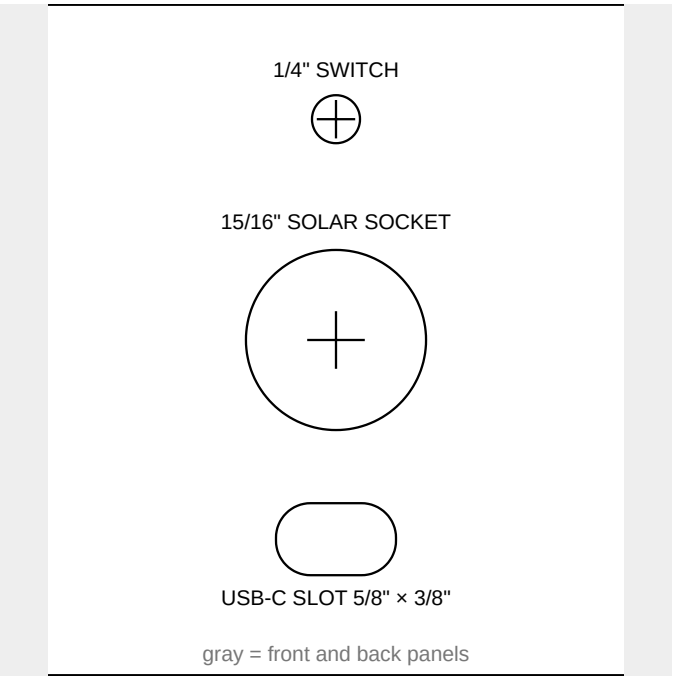
FRONT PANEL · FULL-SIZE TEMPLATE (print at 100% / "actual size")

Check: the box below must measure exactly 10" × 5". Tape it to the plywood, punch each crosshair center with a nail, then drill.



RIGHT SIDE · FULL-SIZE TEMPLATE (print at 100%)

Check: the square below should measure exactly 3-1/2" × 3-1/2". Tape it to the outside face of one side piece.



Line the charger board's USB-C port up with the slot from the inside. The solar socket's nut tightens from the inside; 3/4" wood is thick, so if the thread is too short, recess the inside of the hole with a 1-1/4" spade bit about 3/8" deep.